Ground shakes: what this analysis adds

ALREADY IN THE EDMS SLIDES

The charge precursor and electrical pickup

A slow charge change precedes the main shake. Disconnected light-readout inputs also respond.

ADDED HERE

Tests of how the early pulses relate to charge

The first pulse does not set a fixed timer. A timing peak does not establish a charge–light match.

STILL OPEN

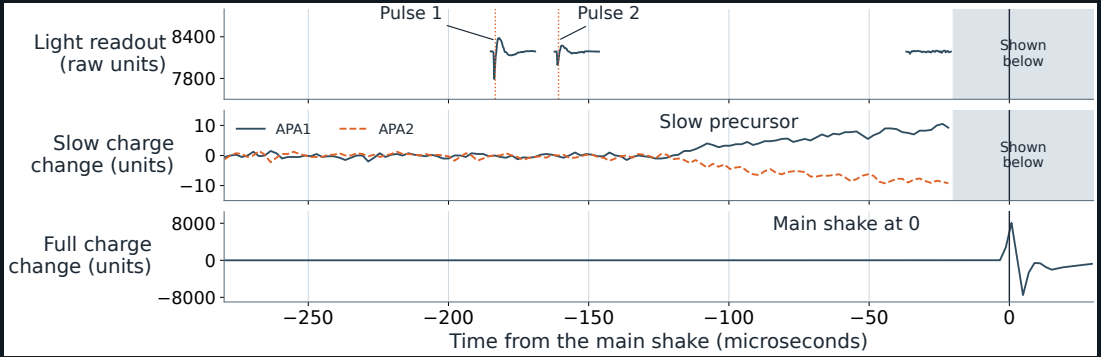
What starts the disturbance

We have added measurements and comparison tests. We have not identified the cause or confirmed an optical flash.

Known observations: EDMS 3493627 v1, pp. 9, 21, 23, 79. Seven 2024 runs; 1,531 saved records.

Three signals, three names

One recorded example: run 32731, event 33 · PDS means the light-readout system

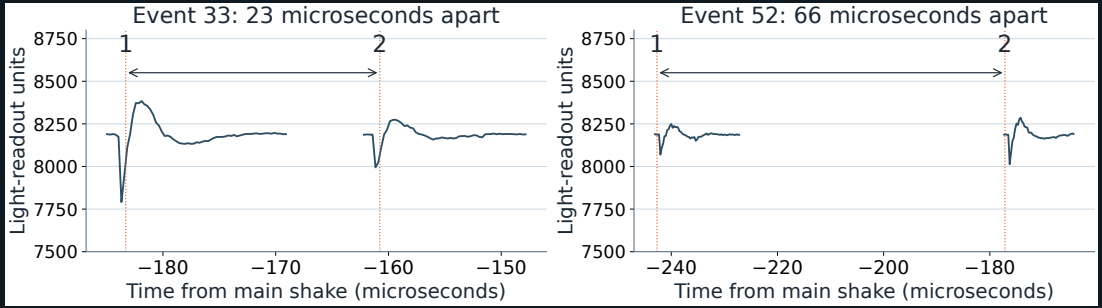


Top: two sharp PDS pulses. **Middle:** the slow charge change, called the precursor. **Bottom:** the much larger main shake at time 0. This shows their order in one record.

Axes use different scales; gaps are unrecorded or excluded. One microsecond = one millionth of a second. Timing alone does not prove cause.

Some records contain two early pulses

Run 32731 · The same connected light-readout input, DAPHNE 109 channel 47

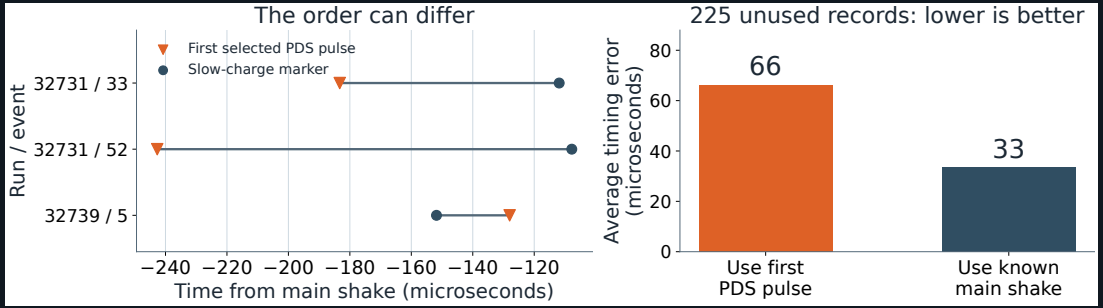


Each dip is a pulse in the recorded signal. The two examples are about 23 and 66 microseconds apart. They do not show a fixed spacing. These early pairs are separate from the later main-shake reflections.

Added count: 101 of 716 records have at least two selected pulses, 100–300 microseconds before the main shake. Pulse candidates are not confirmed light flashes; recording gaps limit this count.

The first pulse does not set a fixed timer

Added test: relate the first selected PDS pulse to the slow charge change

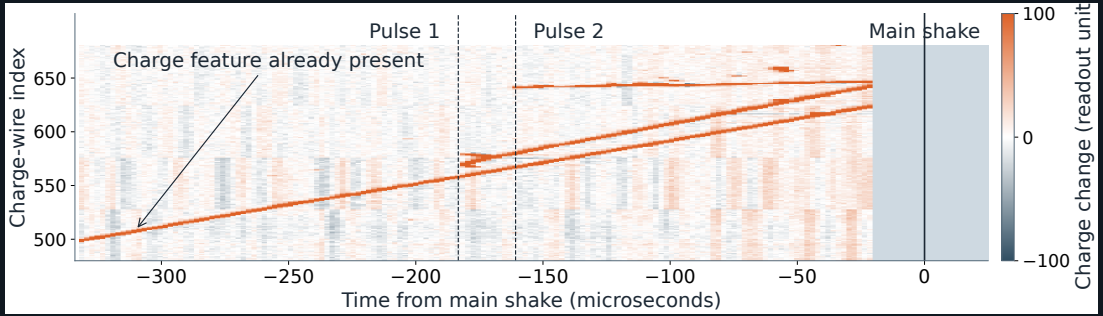


Left: in one example the charge marker comes before the first selected PDS pulse. **Right:** on 225 unused records, timing from that pulse gives twice the error of timing from the known main shake.

The charge marker is a sustained 2-unit change, not an exact physical start. Earlier PDS pulses may be unrecorded. This is a comparison of saved shakes, not an early-warning test.

One proposed charge match fails the time-order check

Added charge–light test · Run 32731, event 33 · Part of the APA2 charge readout

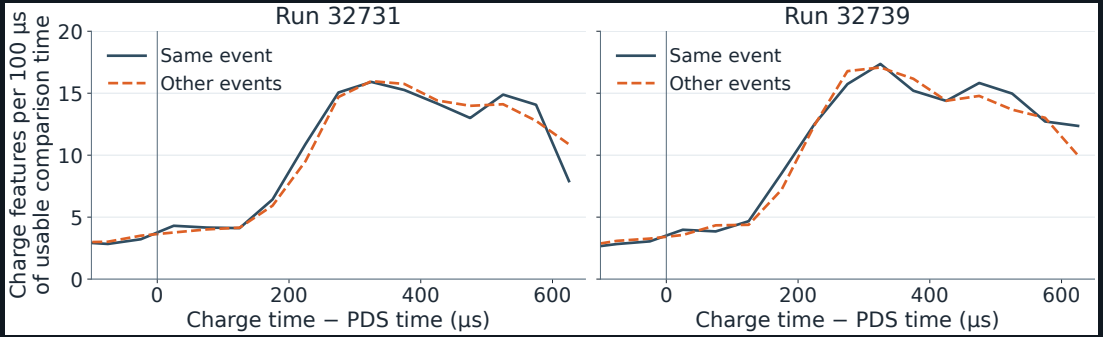


Each row is a charge wire; colour shows its signal. The diagonal feature is already visible before both PDS pulses. Those later pulses cannot mark the start of the whole feature under the recorded timing.

This rejects that particular assignment. Later charge features remain possible matches. Grey means excluded samples. The feature is not a fitted 3D track; electronics delays remain uncalibrated.

The timing peak survives swapping events

Added comparison: keep the charge record; replace its PDS pulses with pulses from other events

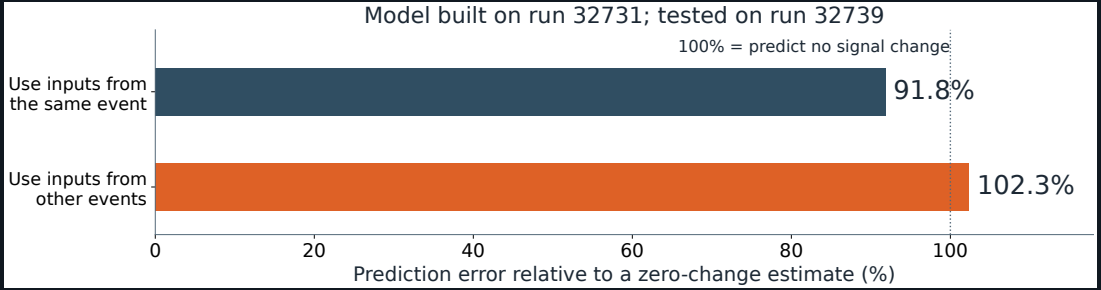


Blue uses pulses from the same event. Orange uses pulses from other events. Both peak near 325 microseconds. The peak therefore does not identify which PDS pulse belongs to a charge feature.

Light from one interaction should precede its drifting charge. This test has not found a unique match. It does not rule out light hidden by the shake or by missing readout.

The new model measures a known electrical effect

Electrical pickup was already documented in EDMS 3493627 v1, pp. 21–23



The model uses reported disconnected inputs to estimate connected-input signals. It performs better with signals from the same event. The addition is a quantitative test on a run that was excluded from fitting.

The prediction is partial and varies by channel. It does not locate the source or separate pure light from electrical pickup. The disconnection state comes from the engineering report.

The slow-control result has a limited scope

Slow controls record operating values such as voltage and current

ADDED TEST

No relationship repeated consistently in both November runs

We compared 32 sufficiently sampled monitors with three waveform measurements. No pair passed the stated repeatability check with one-minute bins.

NOT RETESTED HERE

The EDMS slides report effects of high voltage and wire bias

Our test does not repeat the shake-rate study or the comparison with wire bias removed. It cannot support the claim “high voltage is unrelated to ground shakes”.

A weak current hint depends on time binning and fails the other checks. Two ground monitors are too sparse. Slow samples cannot exclude a brief transient.

Earlier observations: EDMS 3493627 v1, pp. 10, 34–35. New test: runs 32731/32739; selected shake records only. Full plot and criteria are in the backup.

The next tests should answer the open questions

USE EXISTING DATA

Recheck the wire-bias change

Compare the slow PDS signal across the bias ramp in runs 30201 and 30202.

USE EXISTING DATA

Fit whole charge tracks

Compare each complete charge track with the position and timing of its possible light pulse.

ADD THE MISSING COMPARISON

Check similar records without shakes

Use matched settings and measured observation time to check which early pulses distinguish shakes.

Open questions from EDMS 3493627 v1, pp. 34–35, 39, 42. The cause remains unresolved.

Backup: data and definitions

Saved 2024 records

Month	Run	Records
July	27938	6
	27951	19
October	30201	44
	30202	83
November	32730	1
	32731	716
	32739	662
Total		1,531

Charge readout (TPC)

Measures signals from drifting electrons. APA1 and APA2 are the two charge-readout assemblies used here.

Light readout (PDS)

Records photon-sensor signals and electrical interference. A PDS pulse is not automatically a light flash.

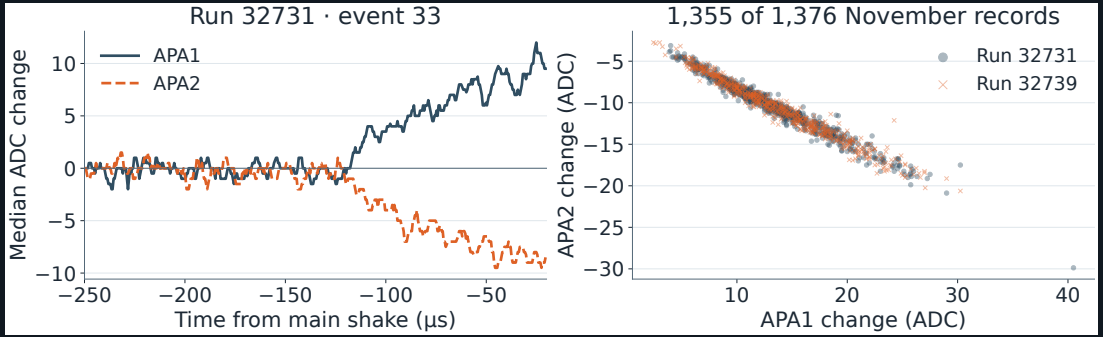
Slow controls (SC)

Operating monitors: 34 channels, 7.74 million archived samples.

Automatic charge screen: 1,516 candidates, five below cut, ten unassessable. Selected files do not measure shake rate. No candidate has a complete 4 ms of pre-shake PDS readout.

Backup: the known charge pattern, counted again

Cross-check of earlier charge observations · EDMS 3493627 v1, pp. 14, 79

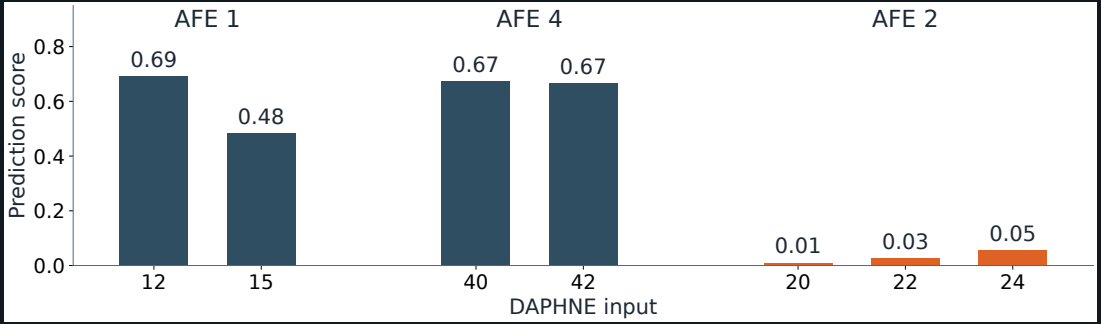


Left: APA1 rises while APA2 falls. **Right:** one dot per record; 1,355 of 1,376 November records have both slow signals clear. This extends the count. The opposite signs were already known.

Median shape agreement after reversing APA2: 0.964 and 0.970. These are readout averages, not reconstructed particle charge or proof of a source.

Backup: the fit depends on the channel

Added numerical comparison · The electronics-dependent pattern was already in EDMS 3493627 v1, p. 21

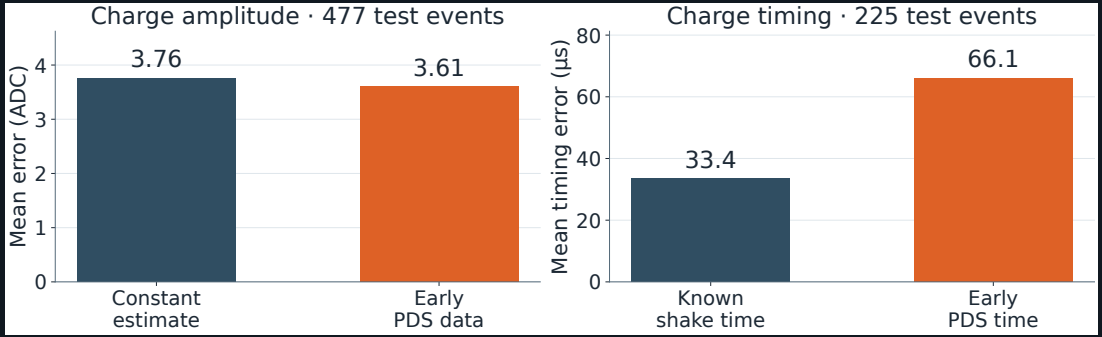


Each bar is a connected input. Taller bars mean better prediction from reported disconnected inputs. Some groups are predicted much better than others.

AFE: an eight-channel front-end chip. Score: 0 = no gain over a zero-change guess; 1 = exact prediction. Trained on run 32731; tested on run 32739. A main-shake-trained fit fails on early pulses.

Backup: early pulses weakly predict charge size

Added test · Fit on run 32731; test on unused run 32739 records

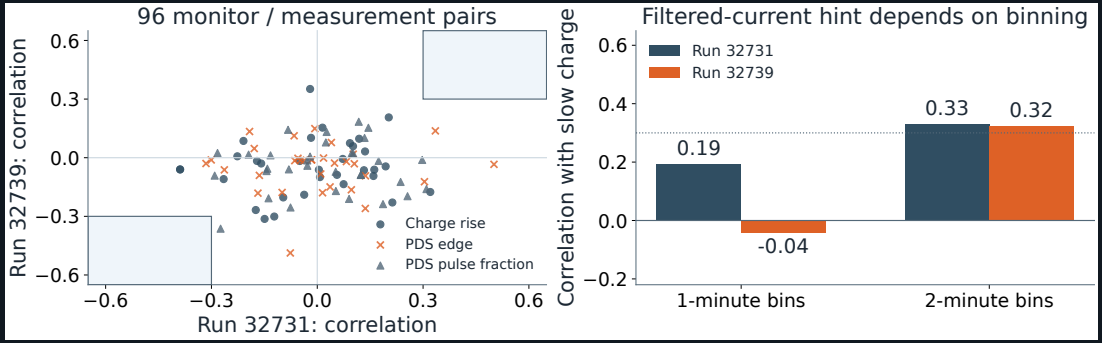


Left: early PDS features reduce the slow-charge size error by only 4.1%. **Right:** the first pulse is a worse clock for the charge marker than the known main shake. Lower bars are better.

The two tests use 477 and 225 records, respectively. Signal size is in raw ADC units. These tests do not measure whether a pulse predicts that a shake will happen.

Backup: what the SC comparison actually tests

32 monitors × 3 measurements · Two November runs · Two further ground monitors too sparse



Left: each dot compares one monitor with one waveform measurement. No dot enters the shaded same-sign repeatability regions with one-minute bins. **Right:** the current hint changes with binning.

Magnitude at least 0.30 in both runs, after file and time-trend adjustment. The two-minute hint also fails timestamp-envelope checks. This does not test shake rate or the bias-off contrast.